

PRACTICAL WEEK-2

Session-2

Task1: Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

```
GNU nano 8.7.1 file_copy.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    int source_fd, dest_fd;
    char buffer[1024];
    ssize_t bytes_read, bytes_written;

    source_fd = open("source.txt", O_RDONLY);

    if (source_fd == -1) {
        perror("Error opening source file");
        return 1;
    }

    dest_fd = open("destination.txt",
                  O_WRONLY | O_CREAT | O_TRUNC,
                  0644);

    if (dest_fd == -1) {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    while ((bytes_read = read(source_fd, buffer, sizeof(buffer))) > 0) {
        bytes_written = write(dest_fd, buffer, bytes_read);

        if (bytes_written != bytes_read) {
            perror("Error writing to destination file");
            close(source_fd);
            close(dest_fd);
            return 1;
        }
    }

    if (bytes_read == -1) {
        perror("Error reading source file");
    }

    close(source_fd);
    close(dest_fd);

    printf("File copied successfully.\n");

    return 0;
}
```

```

Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP$ mkdir -p practical/week2/task1
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP$ cd practical/week2/task1
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week2/task1$ pwd
/home/Priyavallika/OSSP/practical/week2/task1
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week2/task1$ ls
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week2/task1$ nano file_copy.c
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week2/task1$ echo "This is a sample file
for demonstrating Linux system calls." > source.txt
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week2/task1$ cat source.txt
This is a sample file for demonstratting Linux system calls.
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week2/task1$ gcc file_copy.c -o file_copy
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week2/task1$ ls
file_copy  file_copy.c  source.txt
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week2/task1$ ./file_copy
File copied successfully.
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week2/task1$ cat destination.txt
This is a sample file for demonstrating Linux system calls.
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week2/task1$ ls -l
total 28
-rwxr-xr-x 1 Priyavallika Priyavallika 16216 Aug  9 16:34 file_copy
-rw-r--r-- 1 Priyavallika Priyavallika 1082 Aug  9 16:33 file_copy.c
-rw-r--r-- 1 Priyavallika Priyavallika 60 Aug  9 16:34 destination.txt
-rw-r--r-- 1 Priyavallika Priyavallika 60 Aug  9 16:33 source.txt
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week2/task1$ █

```

Observation: The program successfully copied the contents of source.txt to destination.txt using the open(), read(), write(), and close() system calls. The contents of both files were verified to be identical. During execution, the user-space program requested file operations from the Linux kernel through system calls, and the kernel performed the required file operations before returning control to the user-space program.

Task2: Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved.

[illegible]

Observation: The strace utility successfully traced the system calls generated by the cat sample.txt command. The trace showed openat() being used to open sample.txt, followed by file and terminal operations using splice(), pipe(), and fcntl(). The close() system call was used to close the file descriptors, and exit_group(0) indicated successful program termination. The observed system calls demonstrate how the cat command interacts with Linux kernel services to access and display file contents.